

Feasibility Memo (Outline) — ED Arrivals Triage Dataset

To: Mercer ED Board

From: Vishwesh Pattanaik, Clinical AI and Innovation Unit

Re: Is the de-identified ED arrivals dataset good enough to build a triage model from?

Dataset: yaleemmlc_admissionprediction_triage.csv — 55,121 patients, 226 columns

1. (a) Verdict (one sentence)

The dataset is large, complete and clinically rich, and is good enough to build a baseline triage model, provided its single-site, pre-processed nature and class imbalance are documented and respected.

2. (b) Dataset summary

- 55,121 ED arrivals, single US academic emergency department.
- 226 columns: 26 core (demographics, triage vitals, arrival, disposition) and 200 one-hot chief-complaint flags.
- Target: esi, the Emergency Severity Index (1 most urgent to 5 least). No missing values.
- ESI 3, the middle band, is 49% of arrivals (27,010 patients). ESI 2 is 33%, ESI 4 is 16%, ESI 5 is 2%, ESI 1 is 0.1%.
- Vitals present for every patient and within plausible ranges. Cohort is ~58% female, majority White then Black then Hispanic, mostly Medicaid or Medicare.
- Supporting plots: ESI distribution, completeness by group, vital-sign distributions, demographics, chief-complaint frequency, correlations with ESI.

3. (c) Top three quality concerns

4. **Pre-processed, not raw.** Every column is 100% complete and no vitals are implausible. This is the processed Yale release: vitals were imputed and complaints one-hot-encoded upstream. We cannot audit the original missingness or the imputation choices, and this cleanliness will not transfer to Mercer's own extract.
5. **Severe class imbalance.** ESI 1 has only 77 cases and ESI 5 only 1,214. Reliable prediction of the extremes is unlikely; per-class metrics, not overall accuracy, must be reported.
6. **Sparse, high-dimensional complaint features.** 49 of the 200 chief-complaint flags have fewer than 50 positive cases. Left unmanaged, these risk overfitting and should be grouped or dropped.

7. (d) Top three reasons to proceed

1. **Scale and completeness.** 55,121 labelled patients with no missing target is more than enough for a credible baseline.
2. **The target matches the project.** The middle band (ESI 3) is nearly half the data, so there is ample signal exactly where the project is aimed.
3. **Clinical face validity.** Correlations run in the right direction: chest pain, shortness of breath and altered mental status pull toward higher acuity; sore throat and dental pain toward lower. The data behaves as a clinician would expect.

8. (e) Caveats

- Single US academic ED; case-mix and access differ from Mercer and from a Caribbean population. A model trained here needs local re-validation before transfer (Gap 1 of the proposal).
- No single feature is strongly predictive (largest correlation is age at about 0.24). The signal is multivariate.
- Imputation and encoding decisions are inherited and opaque.
- De-identified is not risk-free: the file is git-ignored for size and sensitivity, and only summary statistics leave the environment.